

AI FAQ ASSISTANT API

Solution Architecture

API layers and request sequences

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Layers

Client/Postman -> Express app -> route/validation/auth middleware -> controllers -> Mongoose models/MongoDB. AI controllers call geminiService -> Google Gemini.

Public read sequence

GET request -> FAQ controller/search query -> populate creator display fields -> sort newest-first -> JSON response.

Protected write sequence

Bearer token -> JWT middleware loads req.user -> validation middleware (where configured) -> FAQ controller -> creator ownership check -> save/delete -> JSON response.

Startup

src/server.js loads app, connects to MongoDB, then listens on PORT or 5000. app.js loads dotenv and mounts endpoint groups.

Failure paths

Central error middleware handles request errors, Mongoose validation, invalid IDs, and duplicate keys. Gemini failures are forwarded through the error handler.